

🐸 AMPHIBIANS — STUDY NOTES (BASIC HONOR)



Salamander



Toad



Frog

1. Characteristics of Amphibians

Cold-blooded (body temperature depends on environment)

Vertebrates (have a backbone)

Smooth, moist skin (no scales)

Skin helps with breathing

Lay jelly-like eggs in water

Most live part of life in water and part on land

Undergo metamorphosis

2. Main Orders of Amphibia & How to Distinguish Them

Two main orders commonly taught:

a. Frogs & Toads (Order: Anura)

No tail as adults

Long, powerful hind legs

Jump or hop

Frogs: smooth skin, live near water

Toads: dry, bumpy skin, live more on land

b. Salamanders & Newts (Order: Caudata)

Long bodies with tails

Short legs

Crawl or swim

Moist skin

Usually found in damp places

3. Difference Between Frogs and Toads

Frogs Toads

Smooth, moist skin	Dry, rough, bumpy skin
Live near water	Live farther from water
Long legs for jumping	Shorter legs
Lay eggs in clusters	Lay eggs in strings

4. How Amphibians Protect Themselves

Camouflage (blend into surroundings)

Poison or toxic skin secretions

Warning colors

Jumping or swimming away

Playing dead

Hiding under rocks, logs, or leaves

5. Amphibians Found in a Local Area (Examples)

(Examples — adjust to your country/region)

Frogs — ponds, rivers, gardens

Toads — fields, yards, forests

Salamanders — under logs, near streams

Tree frogs — trees near water

Newts — ponds and wetlands

6. Life History of Amphibians

1. Eggs laid in water
2. Eggs hatch into tadpoles (gills, tail)
3. Tadpoles grow legs
4. Lungs develop
5. Tail shrinks (frogs)
6. Adult amphibian lives on land and water

7. Economic / Environmental Value of Amphibians

Eat insects (mosquito control)

Food for other animals

Used in scientific and medical research

Indicate environmental health

Help balance ecosystems

8. Where Toads Spend Winter or Dry Season

Burrow underground

Hide in mud

Under logs, rocks, or leaf litter

In holes or crevices

Enter hibernation (cold) or aestivation (dry)

9. Identifying Frogs by Sound

Each species has a unique call

Sounds used mainly by males

Calls attract mates and mark territory

Examples:

Croaking

Trilling

Chirping

Ribbit sounds

10. How Frogs and Toads Sing

Air passes over vocal cords

Sound amplified by a vocal sac

Vocal sac inflates like a balloon

Makes sound louder and travel farther

11. Amphibian Observation (General Notes)

Sleep during daytime (most are nocturnal)

Come out at night to feed

Eat insects, worms, small invertebrates

Travel by hopping, crawling, or swimming

Jump distances vary by species

EXTRA AMPHIBIAN NOTES (Helpful for Tests)

Habitat

Ponds

Wetlands

Forests

Streams

Gardens with moisture

Diet

Insects

Worms

Slugs

Small invertebrates

Breathing

Through lungs

Through skin

Tadpoles breathe with gills

Seasonal Behavior

Active in warm, wet seasons

Hide during cold or dry periods

Importance to Humans

Reduce pests

Help farmers

Used in education and research

Environmental indicators

Amphibians – Additional Notes and Study Guide

Examples of Amphibians

Frogs: Live near water, strong jumpers, smooth skin.

Toads: More land-based, bumpy skin, short hops.

Salamanders: Long bodies and tails, live in moist places.

Newts: Semi-aquatic salamanders with rough skin as adults.

Amphibian Habitats

Amphibians live in wetlands, ponds, rivers, forests, gardens, and other moist environments. Clean water is very important for their survival.

Importance of Amphibians

Control insects and pests

Serve as food for other animals

Indicate environmental health

Help scientists study ecosystems

Threats to Amphibians

Pollution and pesticides

Habitat destruction

Climate change

Disease

How Humans Can Help Amphibians

Protect wetlands and ponds

Avoid polluting water sources

Do not harm or collect wild amphibians

Learn and teach others about amphibians

Amphibians – Advanced Honor



Skill Level: 3

Category: Outdoor Nature

Prerequisite

The Amphibians Basic Honor must be completed first.

Study of Amphibians

The scientific study of amphibians is called herpetology.

Amphibian Research and Observation

Amphibians are studied through careful observation in the wild or in captivity without causing harm. Important observations include:

Activity patterns (day or night)

Feeding behavior

Movement and travel distance

Shelter and resting places

Response to temperature and rainfall

Life Cycle and Development

Amphibians undergo metamorphosis:

Egg stage in water

Larval stage (tadpole) with gills

Development of legs and lungs

Adult stage living on land and water

Seasonal Behavior

More active during warm and wet seasons

Enter hibernation in cold seasons

Enter aestivation during dry seasons

Seek shelter underground, in mud, or under logs

Communication and Sound

Mostly males produce sounds

Calls attract mates and defend territory

Sound is produced by vocal cords and amplified by a vocal sac

Different species have unique calls

Environmental Importance

Natural pest control

Part of food chains

Indicators of clean water and healthy ecosystems

Conservation Ethics

Amphibians should not be harmed or collected

Habitats should be protected

Observations should respect wildlife laws and nature

Reptiles – Basic Honor

Reptiles



green sea turtle



crocodile



cobra



chameleon



python



long-necked
tortoise



pricklenape
agama



spiny-tailed
lizard



frilled
lizard



rattlesnake



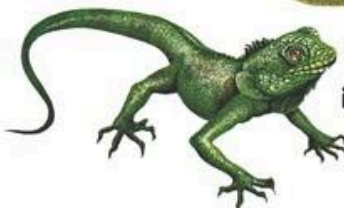
leopard gecko



alligator



flatback
turtle



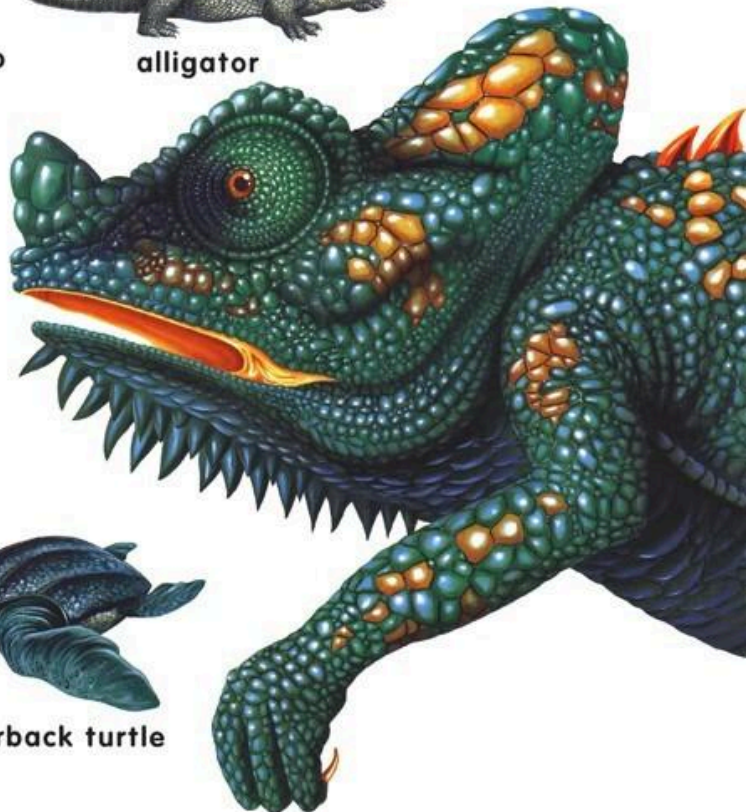
iguana



giant tortoise



leatherback turtle



Skill Level: 1

Category: Nature

What Are Reptiles?

Reptiles are cold-blooded vertebrate animals with dry, scaly skin. They breathe using lungs for their entire lives and usually lay eggs with leathery shells. Reptiles do not go through metamorphosis.

Common reptiles include lizards, snakes, turtles, tortoises, and crocodiles.

Characteristics of Reptiles

Reptiles have scales, lungs, internal fertilization, and are mostly land-dwelling, though some live in water.

Local Reptiles

Reptiles may be found in deserts, forests, grasslands, rivers, and ponds. They help control pests and maintain ecological balance.

Venomous Reptiles

Some reptiles are venomous and use venom to capture prey or defend themselves. Fang type and distribution depend on species.

Snakebite Safety

Venomous bites require immediate medical care. Non-venomous bites should be cleaned and monitored.

Reptiles – Advanced Honor

Reptiles



green sea turtle



crocodile



cobra



chameleon



python



long-necked tortoise



pricklenape agama



spiny-tailed lizard



frilled lizard



rattlesnake



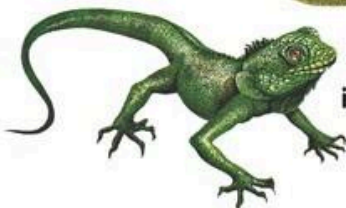
leopard gecko



alligator



flatback turtle



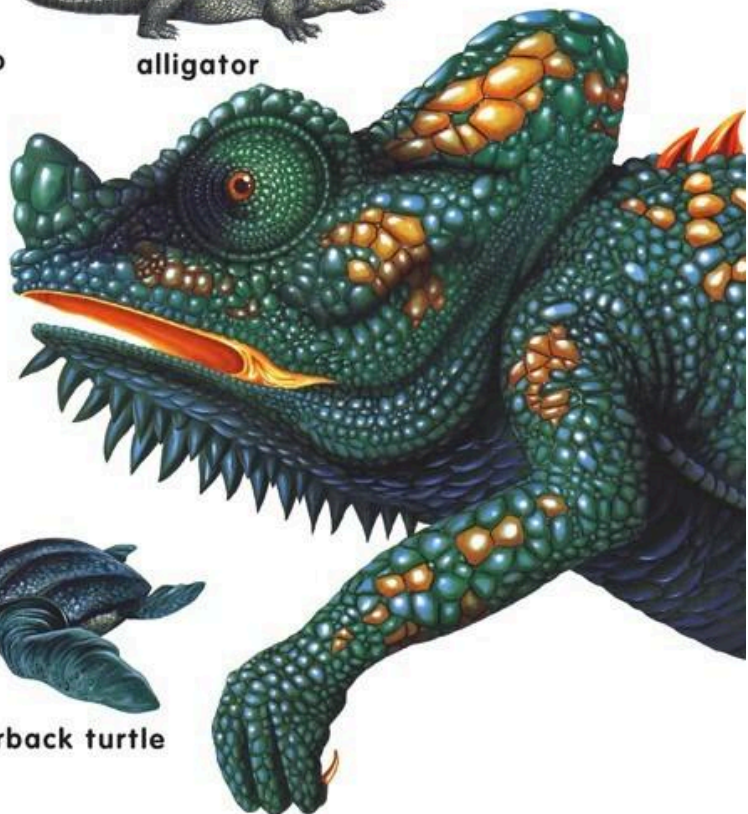
iguana



giant tortoise



leatherback turtle



Skill Level: 3

Category: Nature

Study of Reptiles

The scientific study of reptiles is called herpetology.

Reptiles in Culture

Many folktales exaggerate reptile danger, leading to fear and misunderstanding.

Defense Mechanisms

Reptiles use camouflage, venom, tail dropping, biting, hissing, and protective scales.

Types of Venom

Neurotoxic venom affects the nervous system.

Hemotoxic venom affects blood and tissues.

Field Observation

Reptiles should be observed in the wild, and their behavior carefully recorded.

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